Embedded Controller

USART: Serial Communication Protocol

Young-Keun Kim
School of Mechanical & Control Engineering

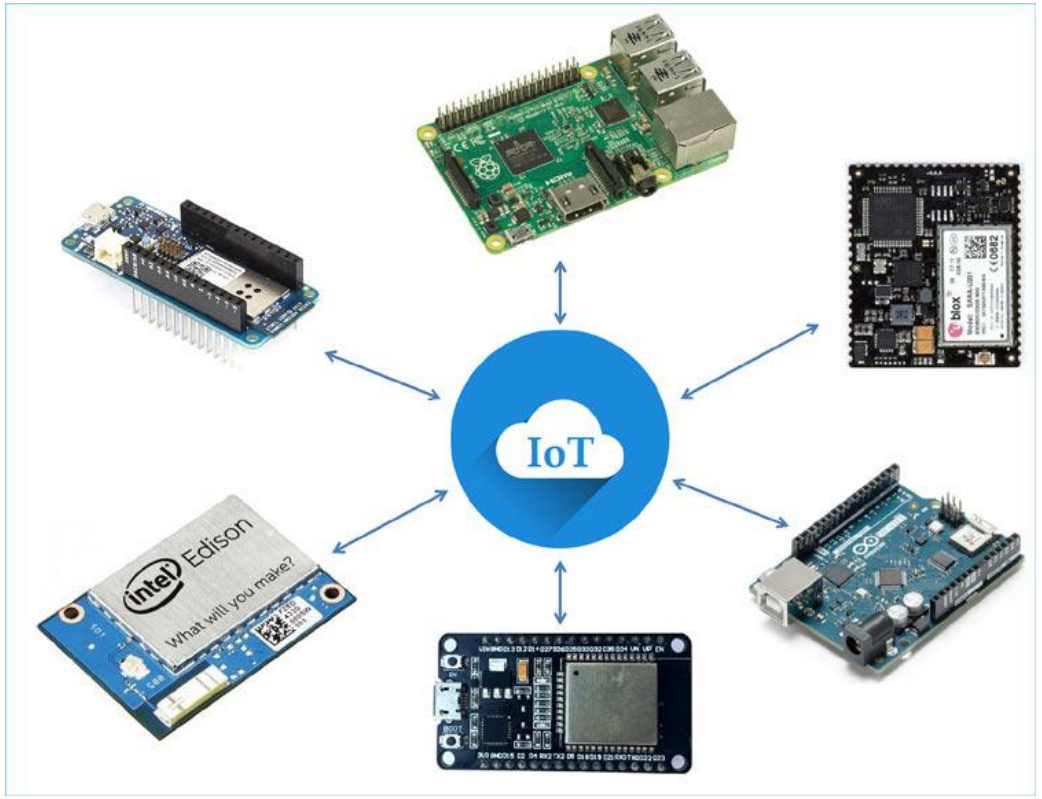
Handong Global University

Introduction: IoT Device

- **Commonly used IoT devices**
  - Microcontroller
  - Single Board Computer(SBC)

- **MCU with IoT**
  - I/O of sensors, actuators and display
  - Network modules
    - Bluetooth, WiFi, LoRA, ZigBee etc

- **Examples of IoT board**
  - Arduino
    - MKR WiFi1010, MKR WAN 1300, Nano 33 IoT
    - Generic Arduino with network shields
  - ESP32/ESP8266 (WiFi)
    - Modules or Boards
  - Raspberry Pi
  - Intel Edison and more



Introduction: IoT Network

- **Network**
  - Network for connecting devices
  - Network for connecting Internet

- **Wired Communication**
  - Serial Communication
  - Ethernet 10/100/1000
  - USB

- **Wireless Communication**
  - Bluetooth
  - WiFi
  - Zigbee
  - LoRA
  - 3G/LTE/5G

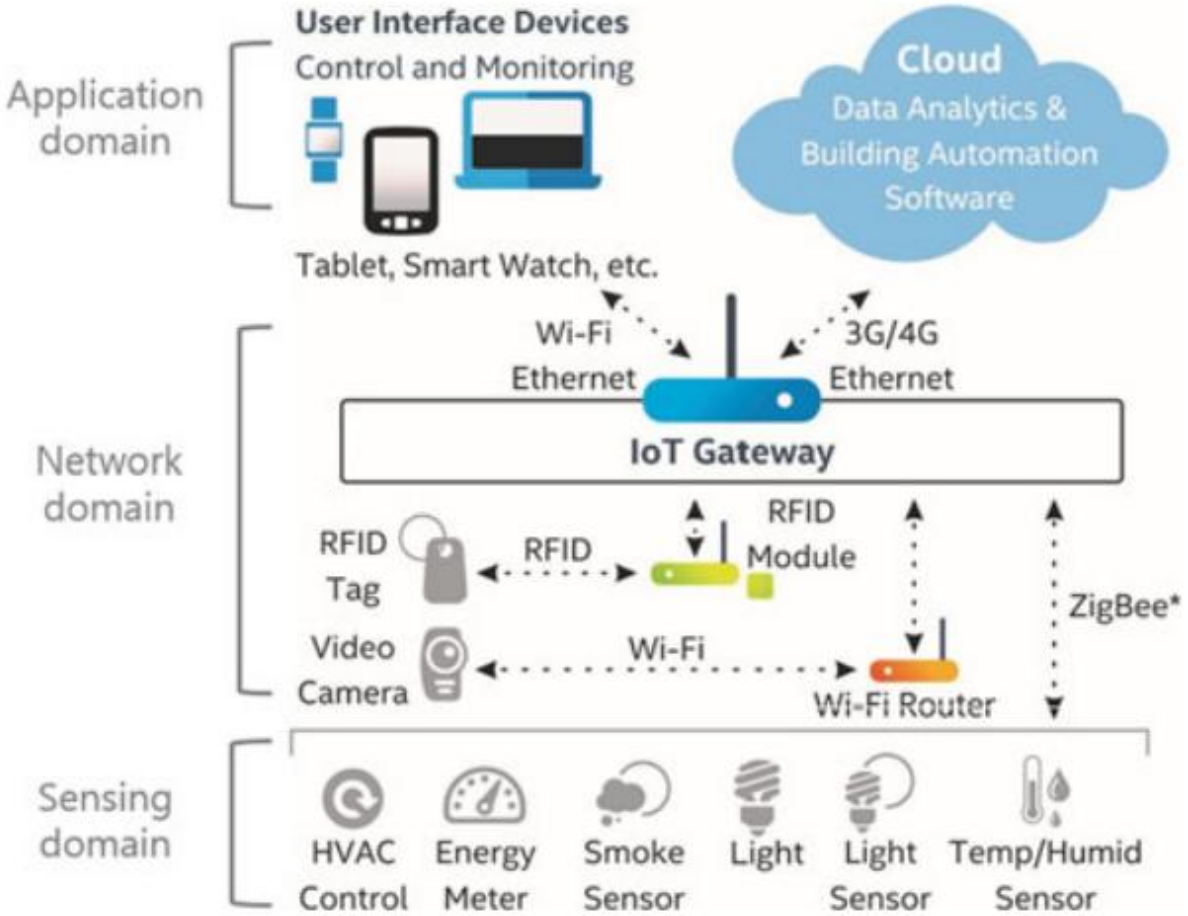


Fig. 1. General feature of IoT gateway.

한동대학교 HANDONG GLOBAL UNIVERSITY

Introduction

- What is communication protocol?
  - A system of description of digital data format and rule for communication

- We will cover protocols of **UART, SPI, I2C**
  - **UART:** Universal Asynchronous Receiver/Transmitter
  - **SPI:** Serial Peripheral Interface (synchronous)
  - **I^2C**: Inter-Integrated Circuit

- Other digital communication protocols/standards
  - Ethernet 10/100/1000
  - Wi-Fi
  - USB 2, 3 and more

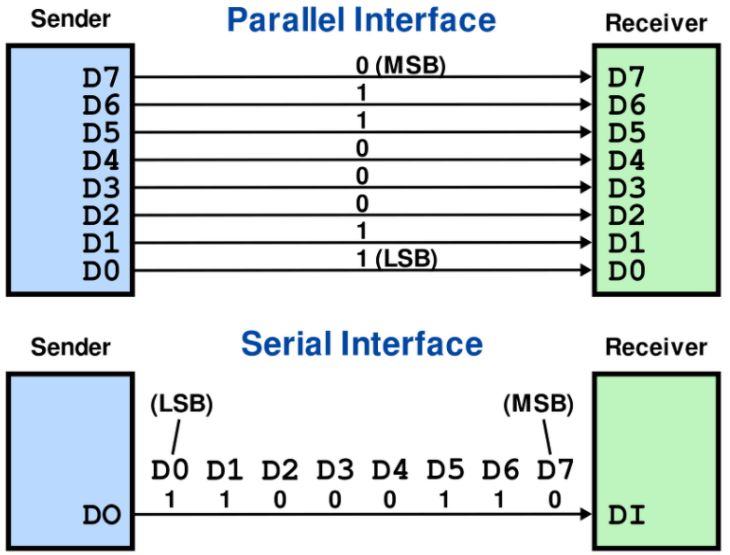
Asynchronous: UART

Introduction

- **How to transmit/receive 8-bit data?**
  - Serial I/O : 1 bit at a time
  - Parallel I/O: multiple bits at once

- **Serial I/O**
  - Fewer wires but fast enough for many application
  - Common Serial Standards
    - **UART(or SCI):** Universal Asynchronous Receiver/Transmitter
      - SCI: Serial Communication Interface

      → RS-232, RS-422, RS-485
    - **SPI:** Serial Peripheral Interface
    - I^2C: Inter-Integrated Circuit
    - Ethernet
    - USB
    - Thunderbolt



* RS: Recommended Standard

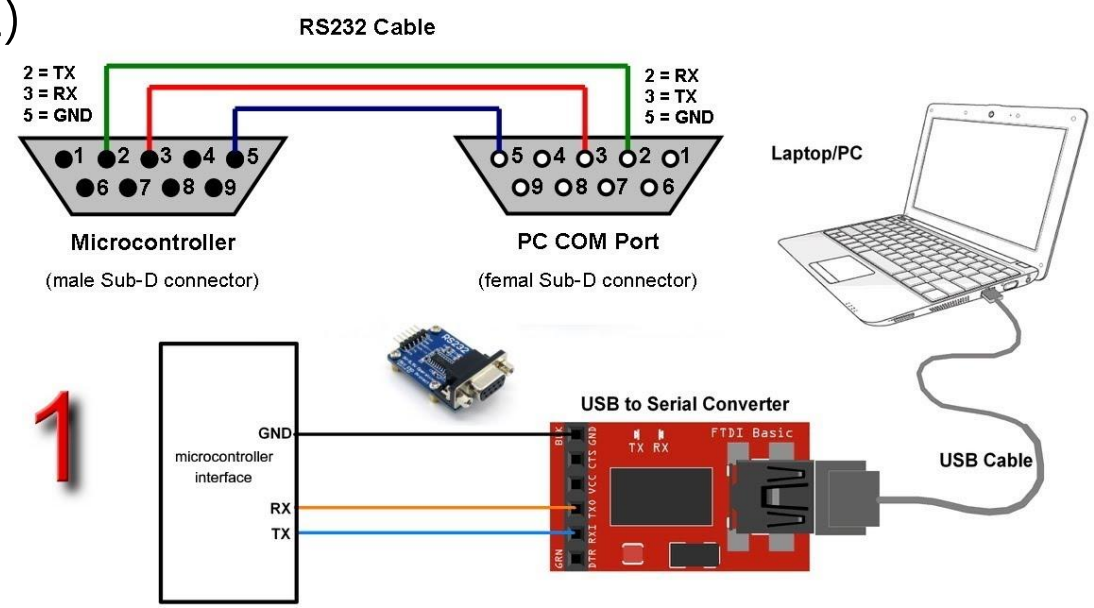
UART Introduction

- **UART(Universal Asynchronous Receiver/Transmitter)**
  - UART itself is not a communication protocol like SPI and I2C
  - **It is a physical circuit** (convert 8-bit parallel to serial data)
  - Initially for Teletypewriter-Modem
  - Later, RS-232 compatible became standards for serial communications for PC
    - It is replaced by USB in modern PC

  - Standards
    - RS-232 (1969) / RS-232C: usually PC
    - RS-422 (apple) / RS-485 (improvement over 422)

    * RS (Recommended Standard)



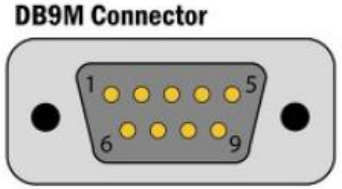


UART Introduction

- **UART(Universal Asynchronous Receiver/Transmitter)**
  - Asynchronous transmission:
    - Serial I/O that communicates without common 'CLOCK'
    - 'Start' and 'Stop' bits instead of CLK.

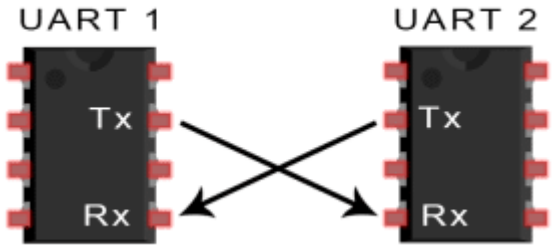
  - Need to agree on transmission rate (Baud Rate). Each device uses its own clock

  - Characteristics
    - Simple to configure, can use two wires, no connected CLOCK
    - Has a parity bit to allow for error checking
    - No Multiple Master/Slave
    - Relatively slow
    - Weak to noise/short range



RS232 Pin Out

Pin #	Signal
1	DCD
2	RX
3	TX
4	DTR
5	GND
6	DSR
7	RTS
8	CTS
9	RI



UART Introduction

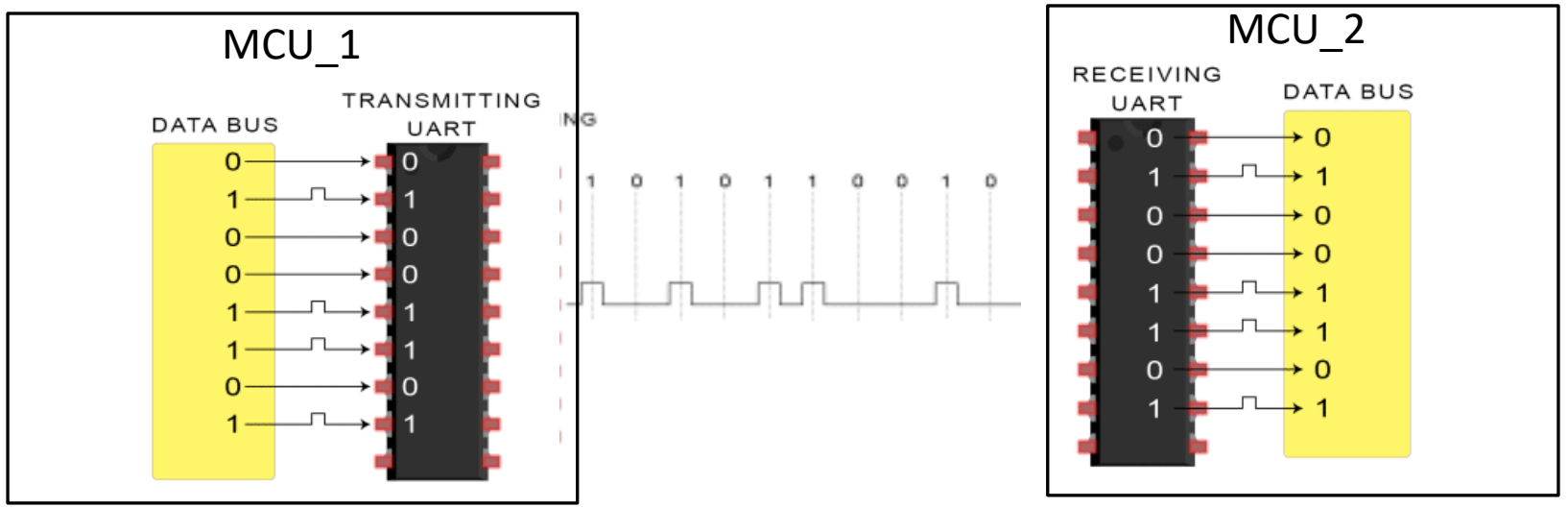
- **UART**
  - Sending a character (data packet) at the given Baud rate
  - **Baud rate:** signaling rate of symbols/second
    - Baud rate: 9600, 14400, 19200, 38400, 57600, and 115200.
    - A symbol is a group of bits. (10 bits for 8bits+start+stop)
    - **Data rate:** bits/seconds (bps)
  - **Start** and **Stop** bits for each 8 bits of data
  - **Parity bit** can be added optionally

Example: 8-bit data(character)+ Start bit+ Stop bit+ no parity. (1 symbol=10bits)
9600 baud rate → 9600 (symbols/s) * (10bits/symbol)= 96,000 bps



UART Introduction

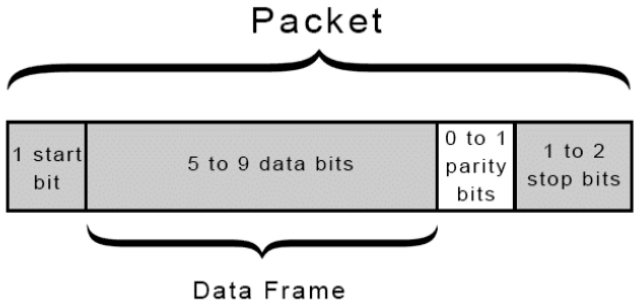
- **UART**
  - Receiver/Transmitter must have the same configuration (protocol)
  - Need to check the voltage level for HIGH (5V, 3.3V etc) and LOW
    - High Level of 3.3V is susceptible to noise signals
  - Data are sent in Parallel to UART Transmitter
  - Add Start(0), Parity, Stop(1)
  - Databus Parallel → UART(TX) → Serial → UART(RX) →Parallel Databus

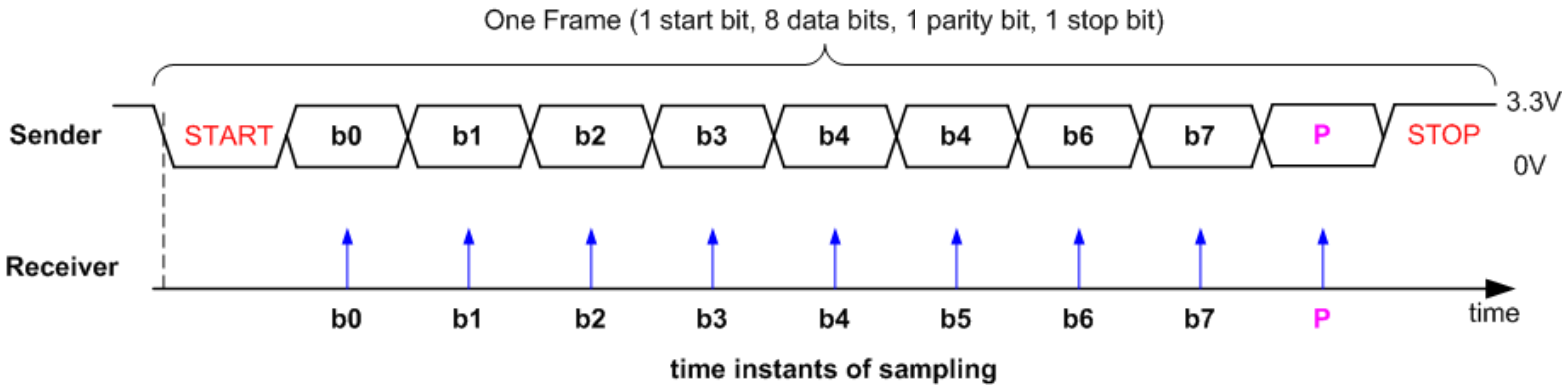


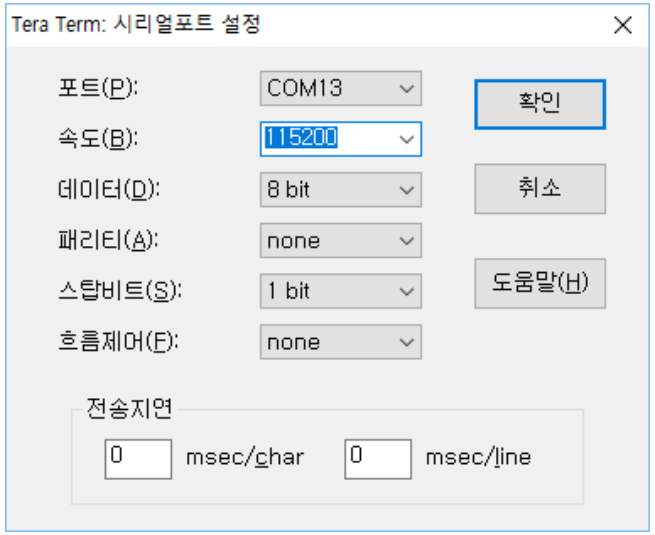
Parallel from data bus | Serial Transmission | Parallel to data bus

UART Introduction

- **UART**
  - Data Packet: Idle(1bit) + Start(1bit) + Data (7~8bits) + Parity(option) + Stop(1, 1.5, 2 bits)
  - Idle: Logic '1' when not in use.
  - Start: Transition '1'->'0' starts transmission
  - Data: Least significant bit first. 5~8bits
  - Parity: Check if transmission is corrupted
    - None/Even/Odd parity
    - Counts even or odd number of logic 'HIGH' bits
      e.g. data= 011 odd parity: parity bit =1
  - Stop bits: Transition '0' → '1' for 1-bit or 2-bits

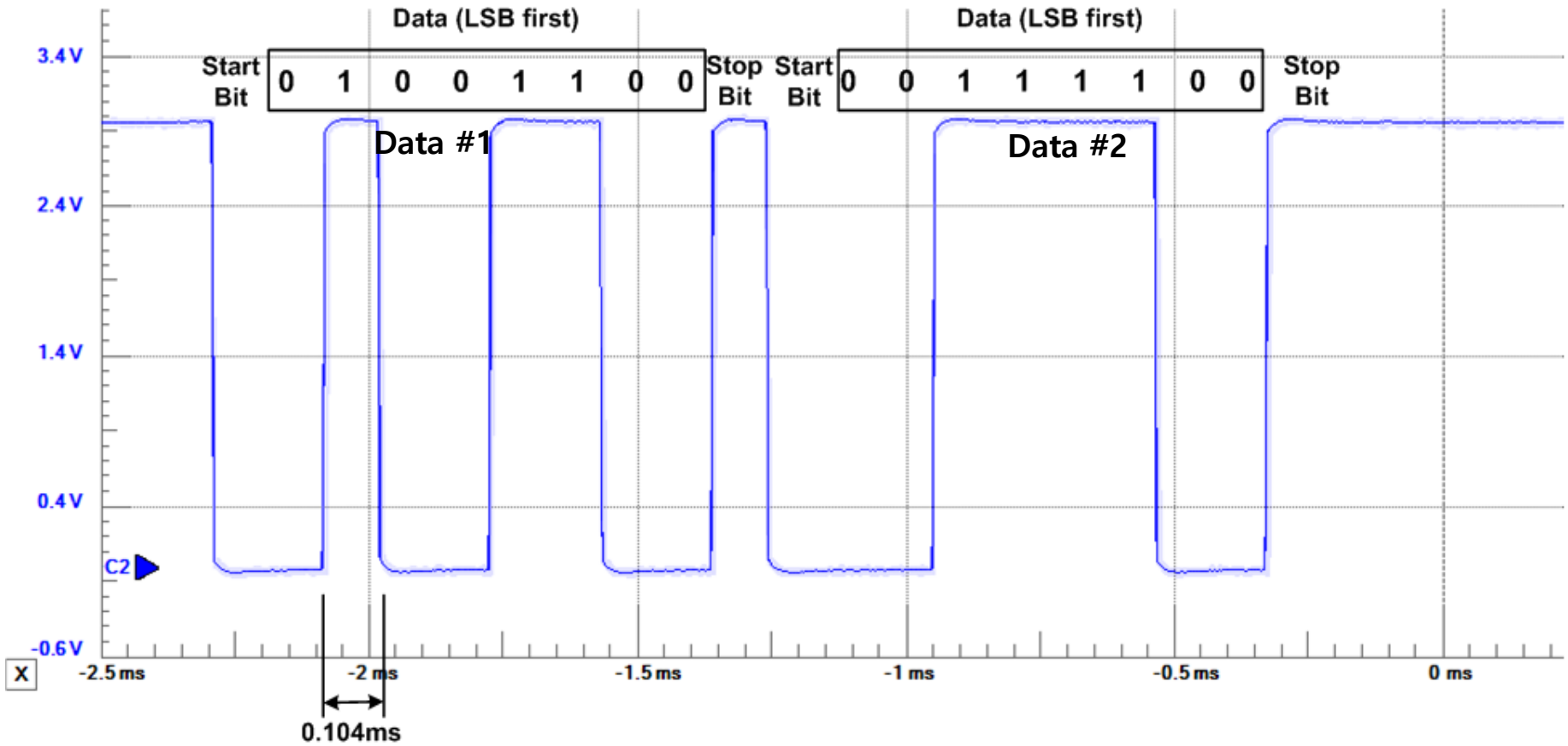






UART Introduction

- Transmission Example:
  - Data #1: Transmitting 0x32 = 0011 0010
  - Data #2: Transmitting 0x3C = 0011 1100



1 start bit, 1 stop bit, 8 data bits, no parity, baud rate = 9600

한동대학교 HANDONG GLOBAL UNIVERSITY

UART Introduction

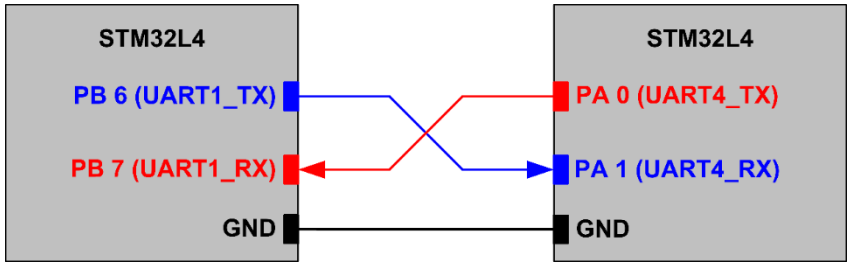
- **Comparison of UART standards**
  - UART has several standards

Characteristics of RS232, RS422, RS423 and RS485

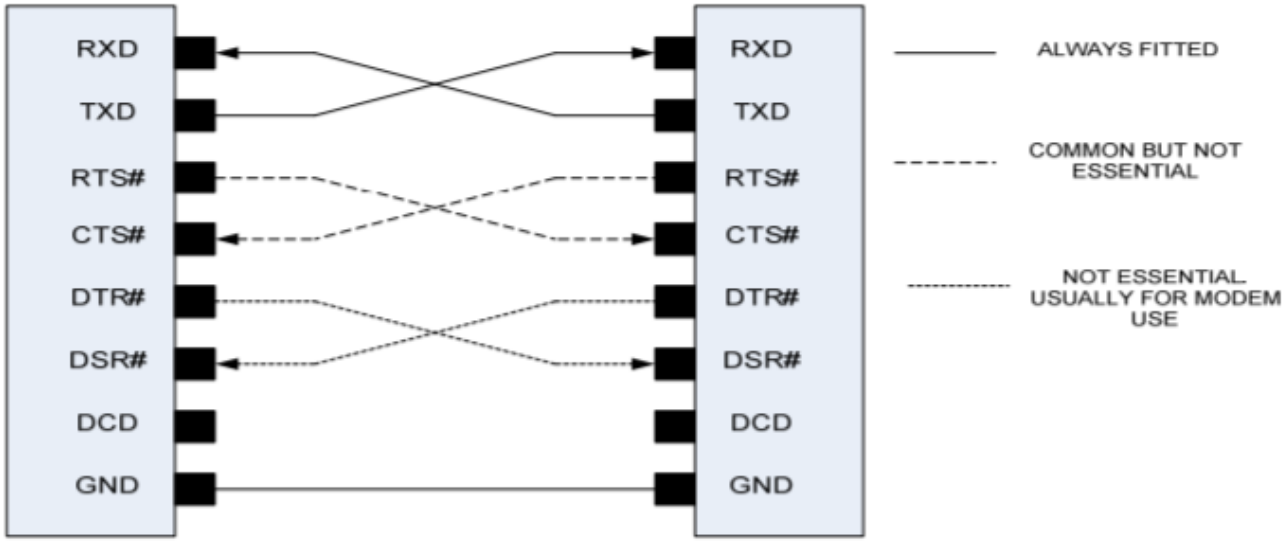
	RS232	RS423	RS422	RS485
Differential	no	no	yes	yes
Max number of drivers Max number of receivers	1 1	1 10	1 10	32 32
Modes of operation	half duplex full duplex	half duplex	half duplex	half duplex
Network topology	point-to-point	multidrop	multidrop	multipoint
Max distance (acc. standard)	15 m	1200 m	1200 m	1200 m
Max speed at 12 m Max speed at 1200 m	20 kbs (1 kbs)	100 kbs 1 kbs	10 Mbs 100 kbs	35 Mbs 100 kbs
Max slew rate	30 V/µs	adjustable	n/a	n/a
Receiver input resistance	3..7 kΩ	≧ 4 kΩ	≧ 4 kΩ	≧ 12 kΩ
Driver load impedance	3..7 kΩ	≧ 450 Ω	100 Ω	54 Ω
Receiver input sensitivity	±3 V	±200 mV	±200 mV	±200 mV
Receiver input range	±15 V	±12 V	±10 V	-7..12 V
Max driver output voltage	±25 V	±6 V	±6 V	-7..12 V
Min driver output voltage (with load)	±5 V	±3.6 V	±2.0 V	±1.5 V

UART Introduction

- **UART: RS-232**
  - MCU typically use simplified 3-pins
  - (GND, TX, RX), and possibly (RTS and CTS)



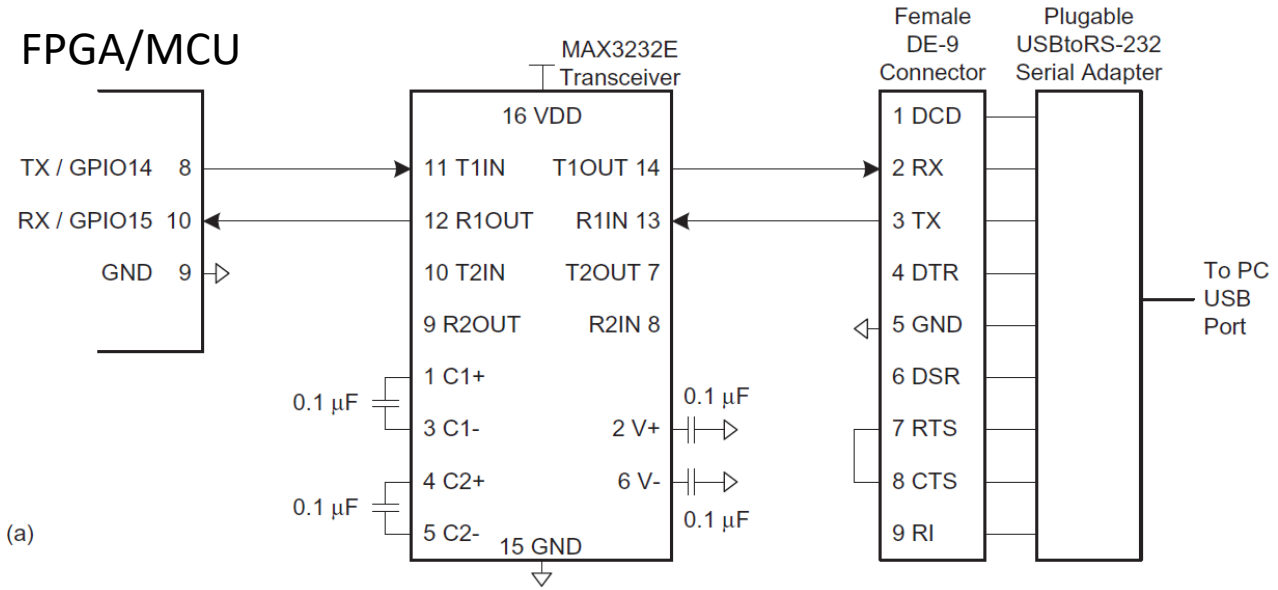
Data	
TXD (pin 3)	Serial Data Output
RXD (pin 2)	Serial Data Input
Handshake	
RTS (pin 7)	Request to Send
CTS (pin 8)	Clear to Send
DSR (pin 6)	Data Set Ready
DCD (pin 1)	Data Carrier Detect
DTR (pin 4)	Data Terminal Ready
Ground	
GND (pin 5)	Ground



DE-9 male cable in PC

UART Introduction

- **Connecting with a Computer USB (RS-232)**
  - Usually for sensing text messages to PC←→ MCU via telnet/serial monitor
  - Typically use **MAX232**: match the same level of signal





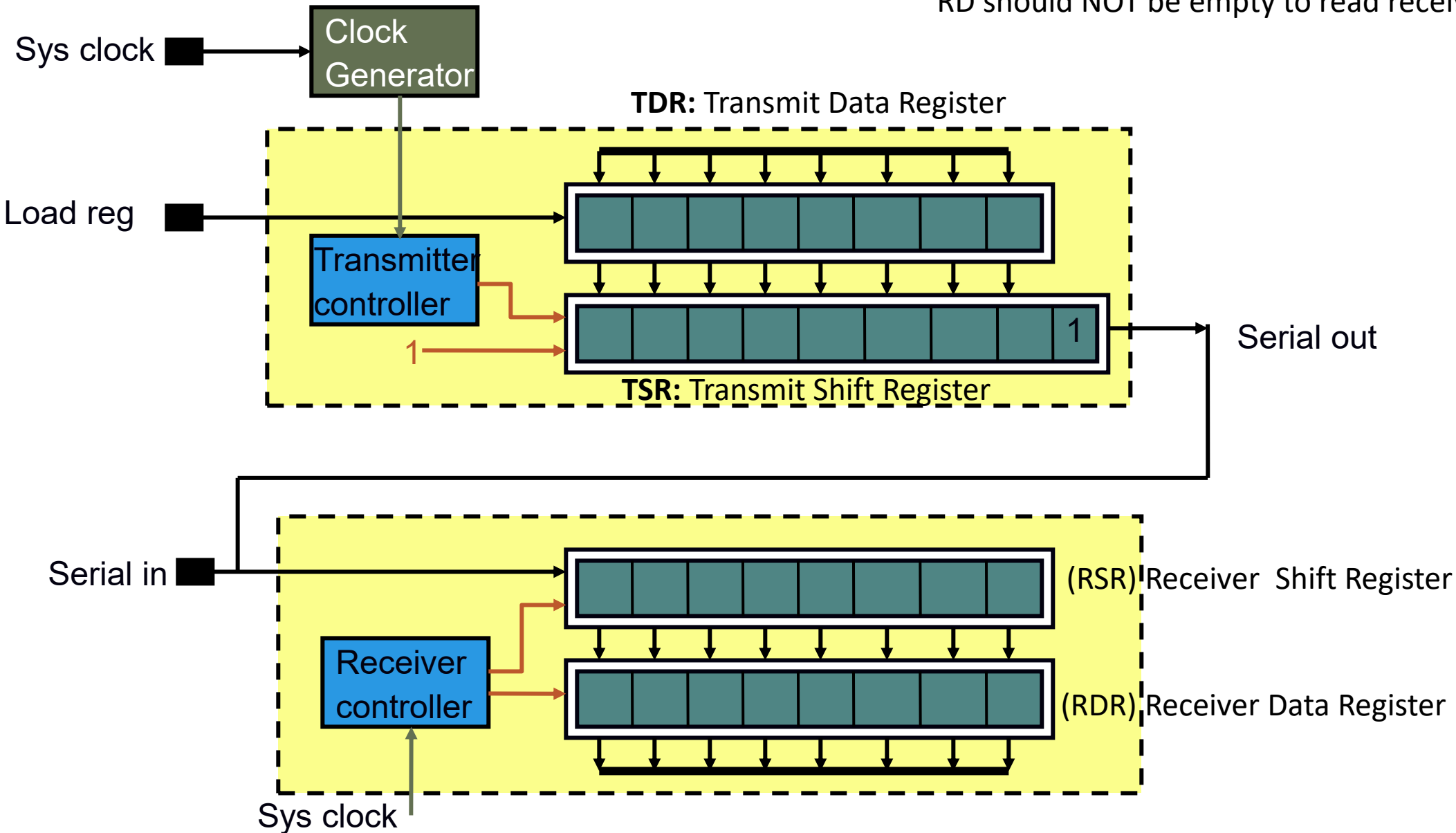
Commonly used ICs

MAX14840E	40Mbps, +3.3V, RS-485 Half-Duplex Transceivers
MAX232	+5V-Powered, Multichannel RS-232 Drivers/Receivers
MAX3162	+3.0V to +5.5V, 1µA, RS-232/RS-485/422 Multiprotocol Transceivers
MAX3232	3.0V to 5.5V, Low-Power, Up to 1Mbps, True RS-232 Transceivers
MAX485	Low-Power, Slew-Rate-Limited RS-485/RS-422 Transceivers

UART Sending and Receiving Data

- **How to implement UART in Digital System**

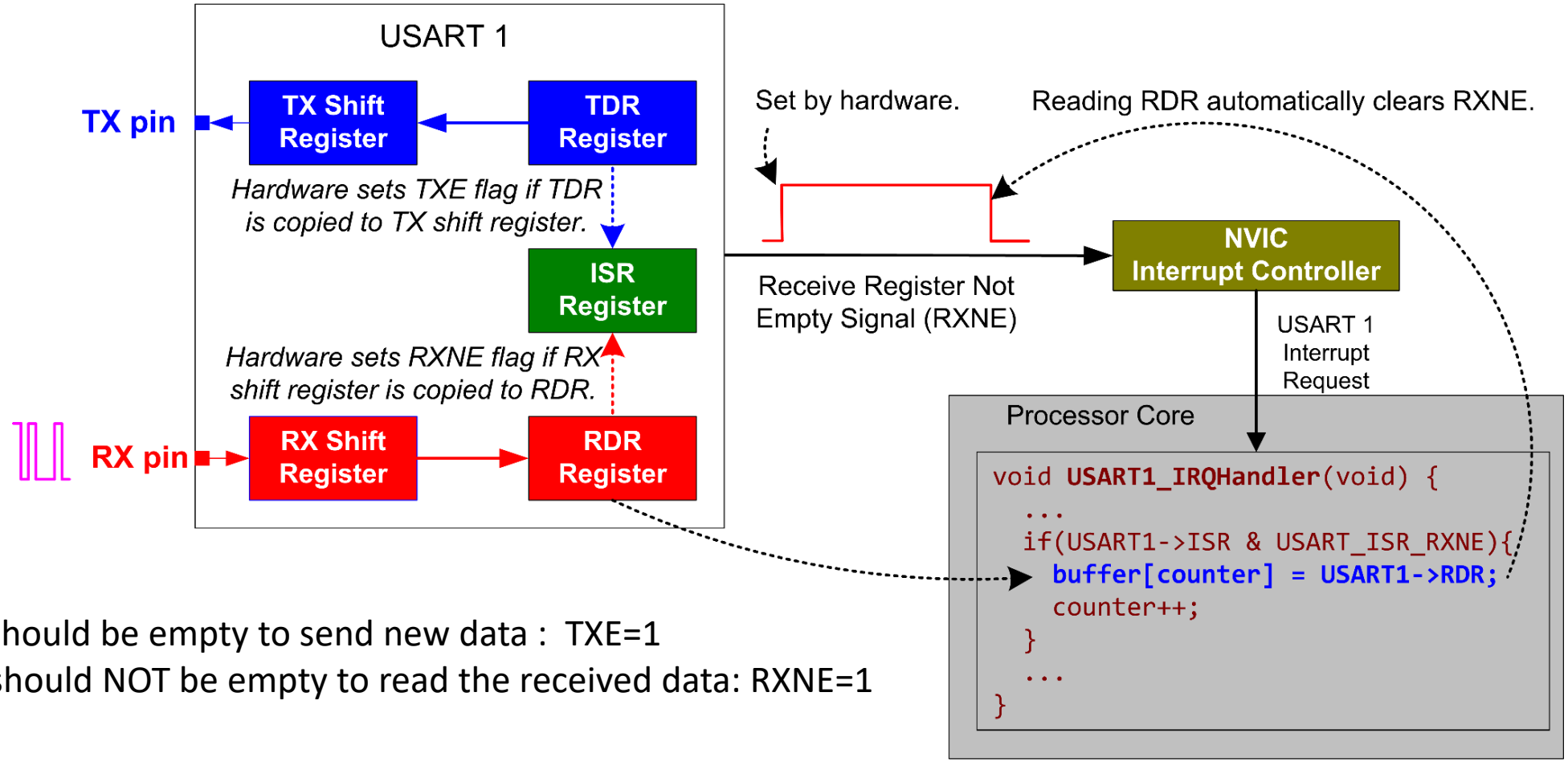
TD should be empty to send data : TXE=1
RD should NOT be empty to read receive data: RXNE=1



UART Interrupt

UART Interrupt

- Polling vs Interrupt. Which is more burden to CPU?
- Usually, RX: Interrupt, TX: Polling



TD should be empty to send new data : TXE=1

RD should NOT be empty to read the received data: RXNE=1

한동대학교 HANDONG GLOBAL UNIVERSITY

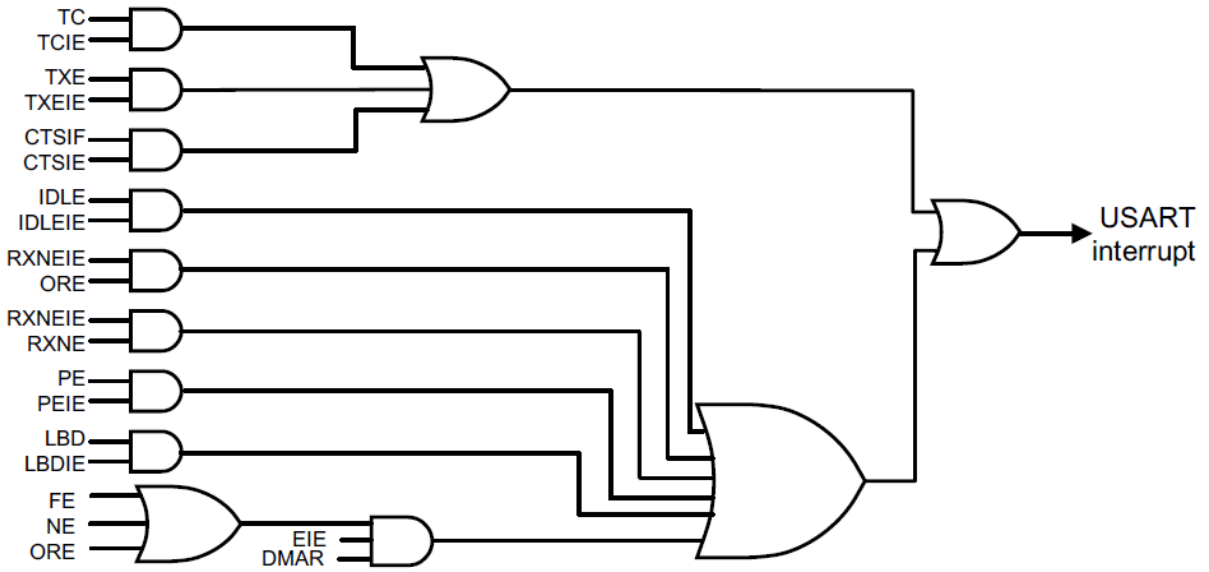
UART Interrupt

- **UART Interrupt**
  - For STM32f4x

Table 86. USART interrupt requests

Interrupt event	Event flag	Enable control bit
Transmit Data Register Empty	TXE	TXEIE
CTS flag	CTS	CTSIE
Transmission Complete	TC	TCIE
Received Data Ready to be Read	RXNE	RXNEIE
Overrun Error Detected	ORE	
Idle Line Detected	IDLE	IDLEIE
Parity Error	PE	PEIE
Break Flag	LBD	LBDIE
Noise Flag, Overrun error and Framing Error in multibuffer communication	NF or ORE or FE	EIE

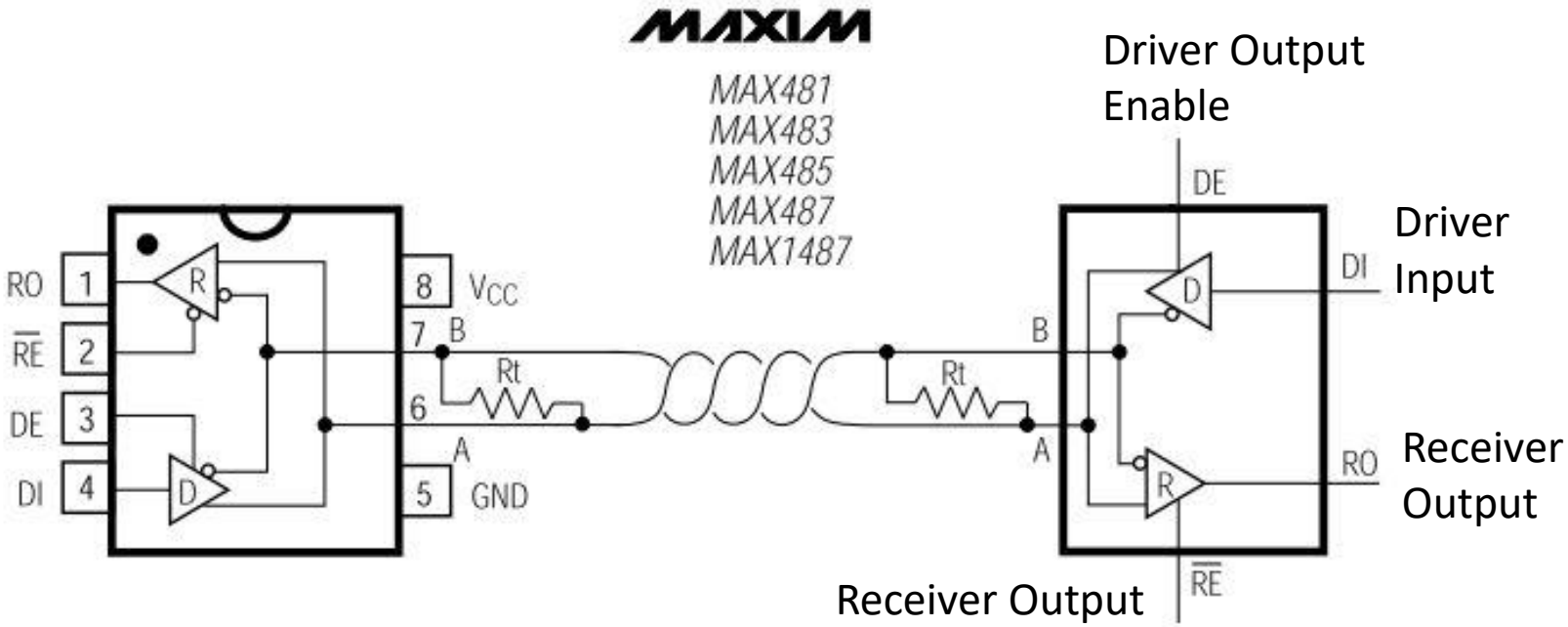
Figure 191. USART interrupt mapping diagram



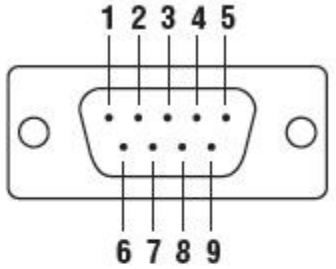
한동대학교 HANDONG GLOBAL UNIVERSITY

UART: RS-485

- **UART: RS-485**
  - Use Differential voltage, more robust to noise RS-232
  - Can allow for greater distances
  - 100~120 ohm for load resistors (Rt)
  - Half-Duplex(two wires)
    - Receive or Send one at a time



DB9 male RS-422/485 port



PIN	RS-422/RS-485-4w	RS-485-2w
1	TxD+(B)	–
2	TxD-(A)	–
3	RxD+(B)	Data+(B)
4	RxD-(A)	Data-(A)
5	GND	GND
6	–	–
7	–	–
8	–	–

한동대학교 HANDONG GLOBAL UNIVERSITY

UART: RS-485

- **UART: RS-485**
  - Can allow up to 10 nodes
  - 100~120 ohm for load resistors (Rt)
  - RS-485 compatible with RS-422. But opposite is not true (Different voltage range)

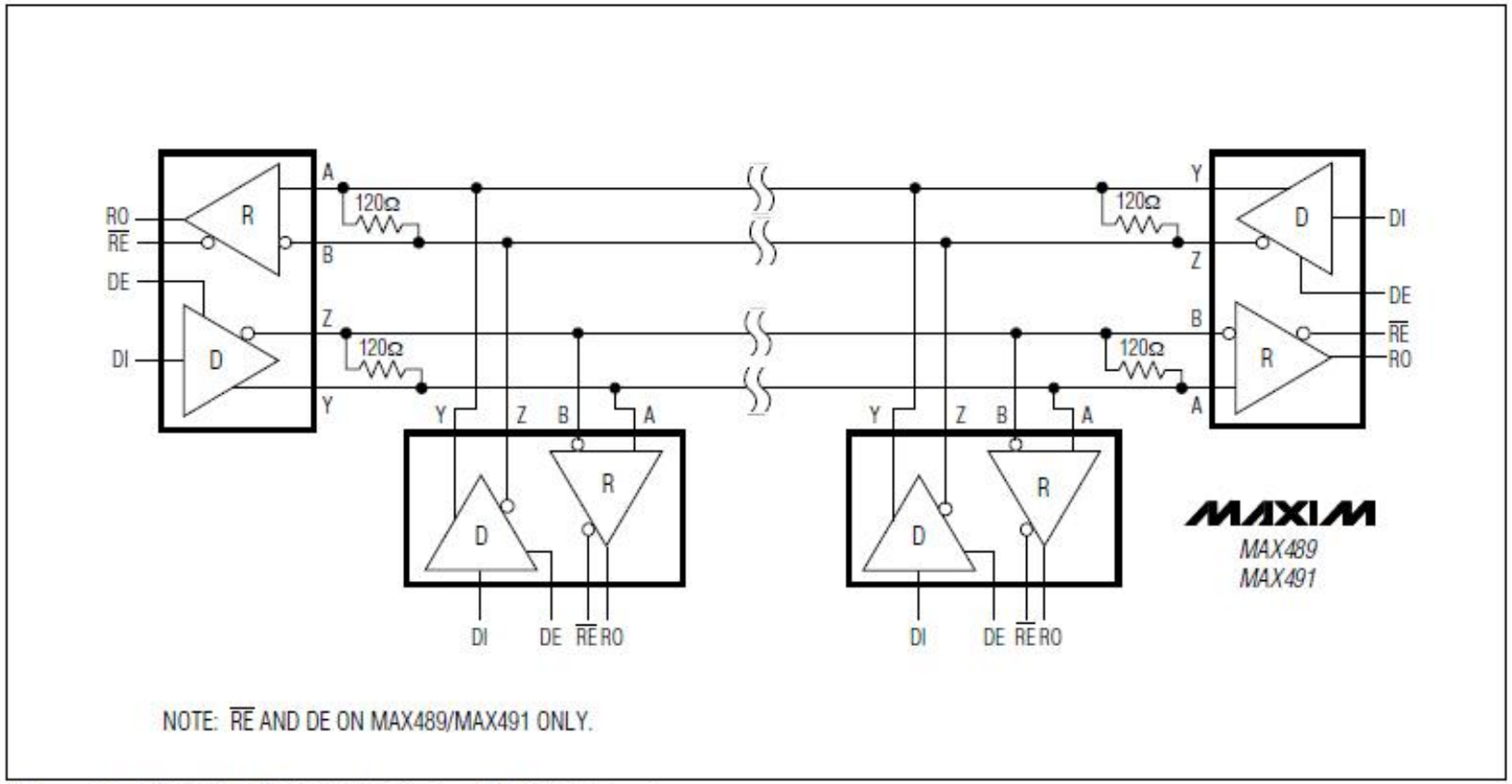


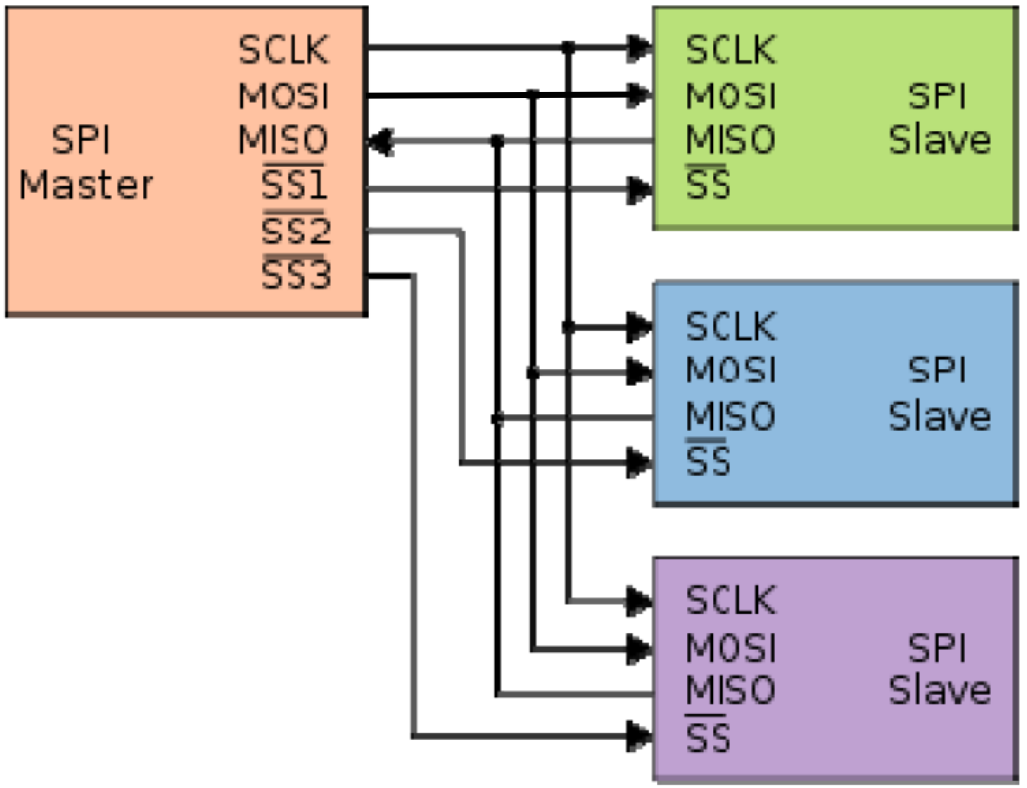
Figure 22. MAX488–MAX491 Full-Duplex RS-485 Network

한동대학교 HANDONG GLOBAL UNIVERSITY

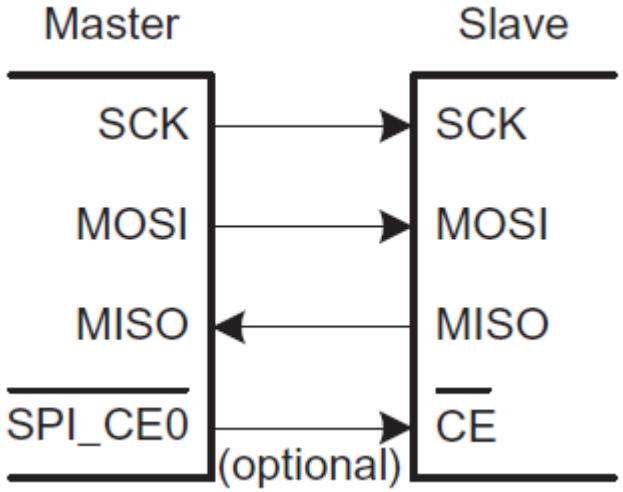
Synchronous: SPI

SPI (Serial Peripheral Interface)

- **SPI**
  - A simple **synchronous** serial protocol
  - Communication between Master and Slave
  - Faster rate than UART (few MHz) and can have multiple Slaves
  - Ports:
    - Serial Clock (SCK),
    - Master Out Slave In (MOSI, or SDO)
    - Master In Slave Out (MISO, also known as SDI).

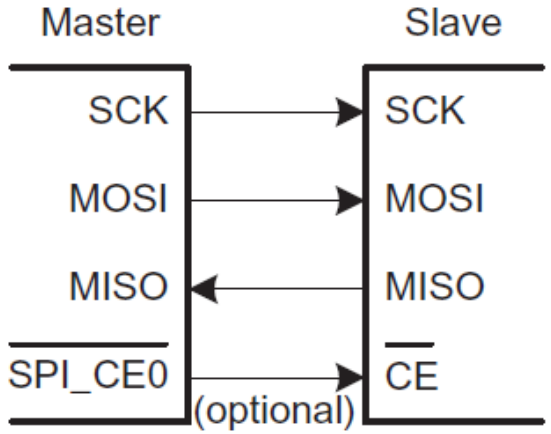


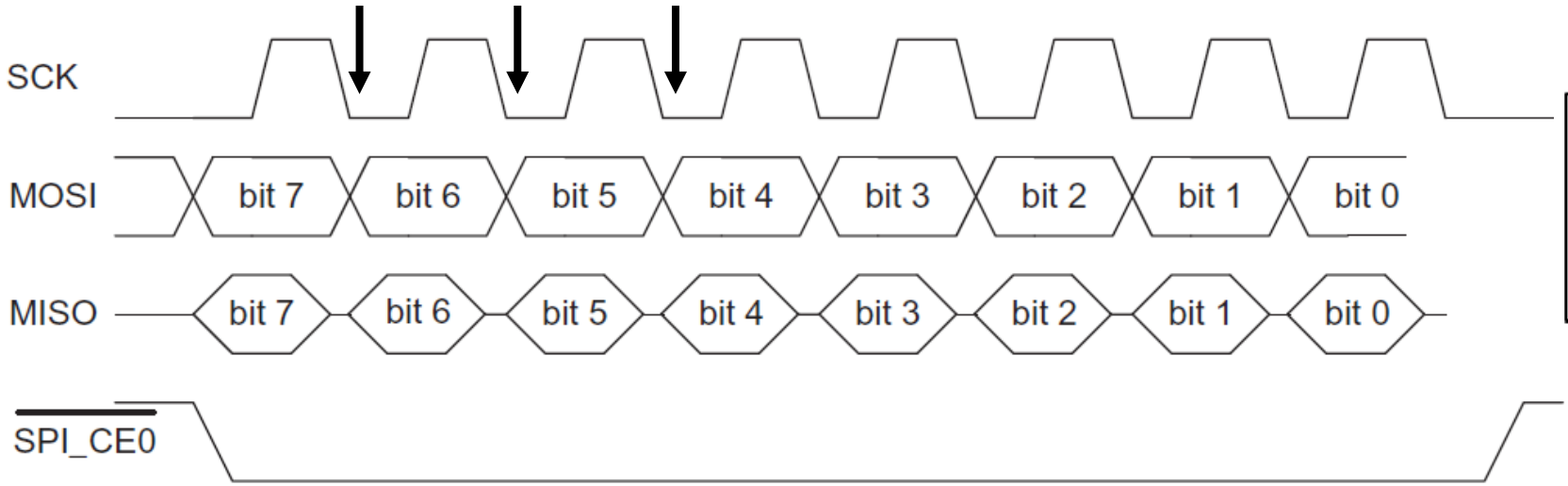
SPI0CLK	SPI Frequency (kHz)
2	125000
4	62500
8	31250
16	15625
32	7812
64	3906

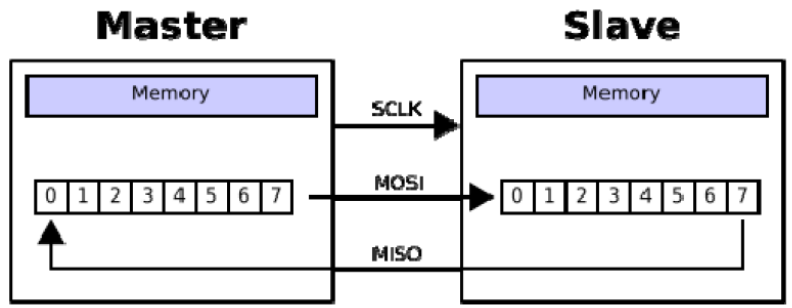


SPI (Serial Peripheral Interface)

- **SPI**
  - The master unit generates the CLK
  - Initiates communication by sending a series of clock pulses on SCK
  - Bits change on the falling edge of SCK
  - Sending data from Master → Slave
    - MSB first, TX data on MOSI
    - Slave may simultaneously respond by putting data on MISO
  - (option) Active-low chip enable: to alert the receiver that data is sent

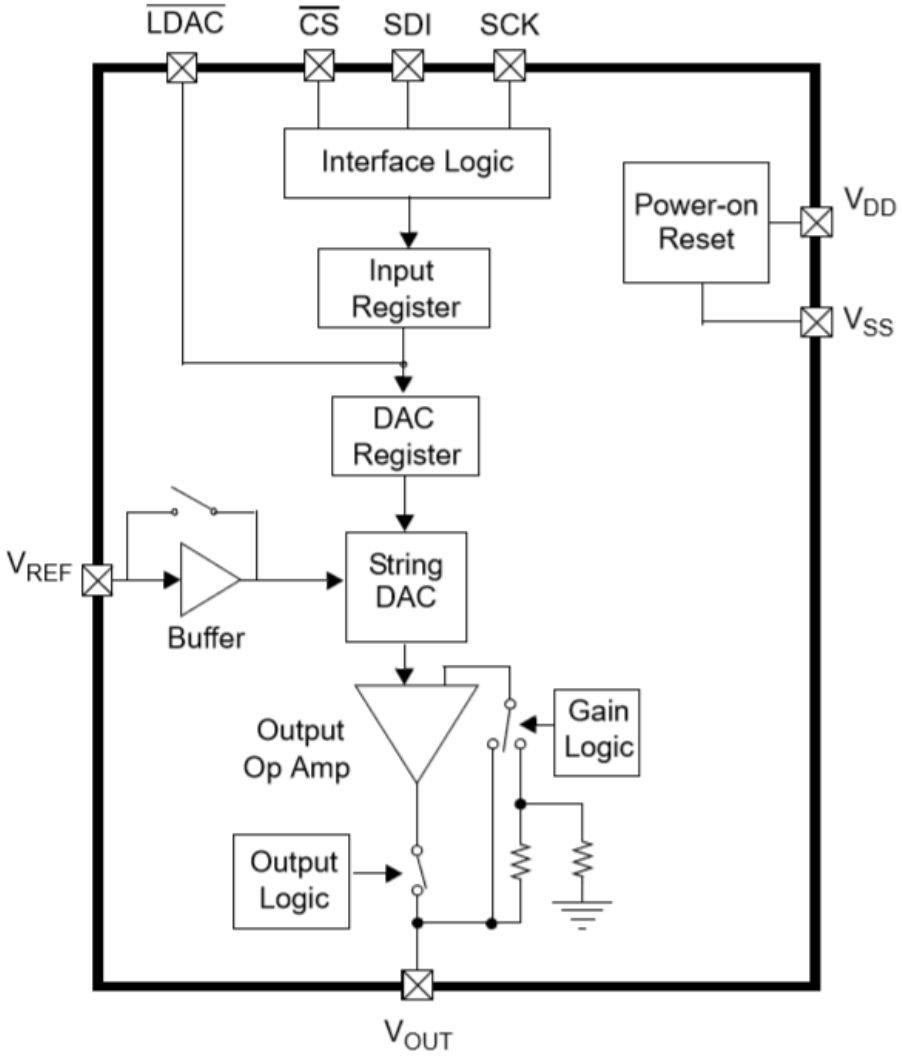






SPI (Serial Peripheral Interface)

SPI example: DAC, Microchip MCP4911





MCP4911 ☆

Status: In Production

View Datasheet

Features:

- 10-bit Resolution

MICROCHIP **MCP4901/4911/4921**

8/10/12-Bit Voltage Output Digital-to-Analog Converter with SPI Interface

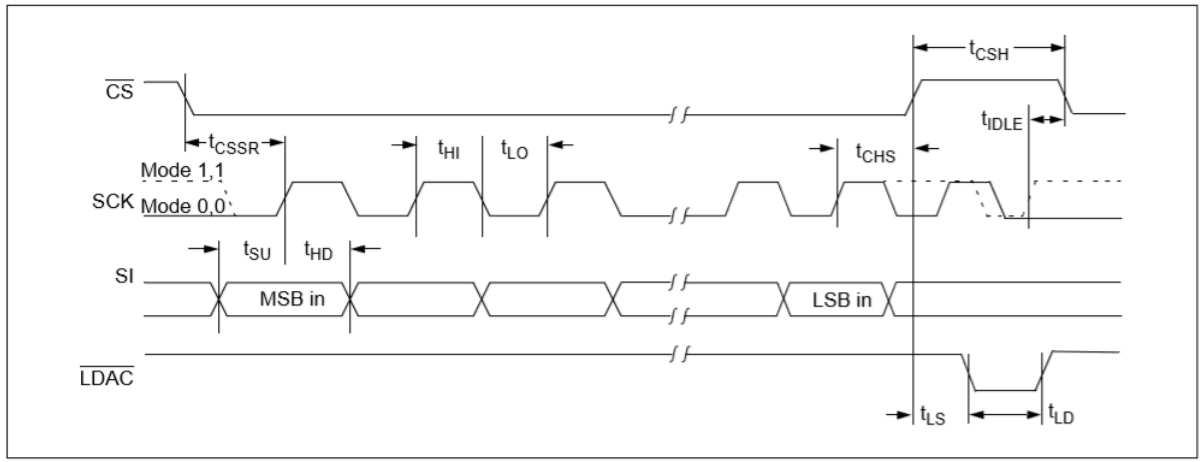


FIGURE 1-1: *SPI Input Timing Data.*

https://www.microchip.com/wwwproducts/en/MCP4911

SPI (Serial Peripheral Interface)

SPI example: DAC, Microchip MCP4911

REGISTER 5-2: WRITE COMMAND REGISTER FOR MCP4911 (10-BIT DAC)

W-x	W-x	W-x	W-0	W-x	W-x	W-x	W-x	W-x	W-x	W-x	W-x	W-x	W-x	W-x	W-x
0	BUF	$\overline{GA}$	$\overline{SHDN}$	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	x	x
bit 15															bit 0

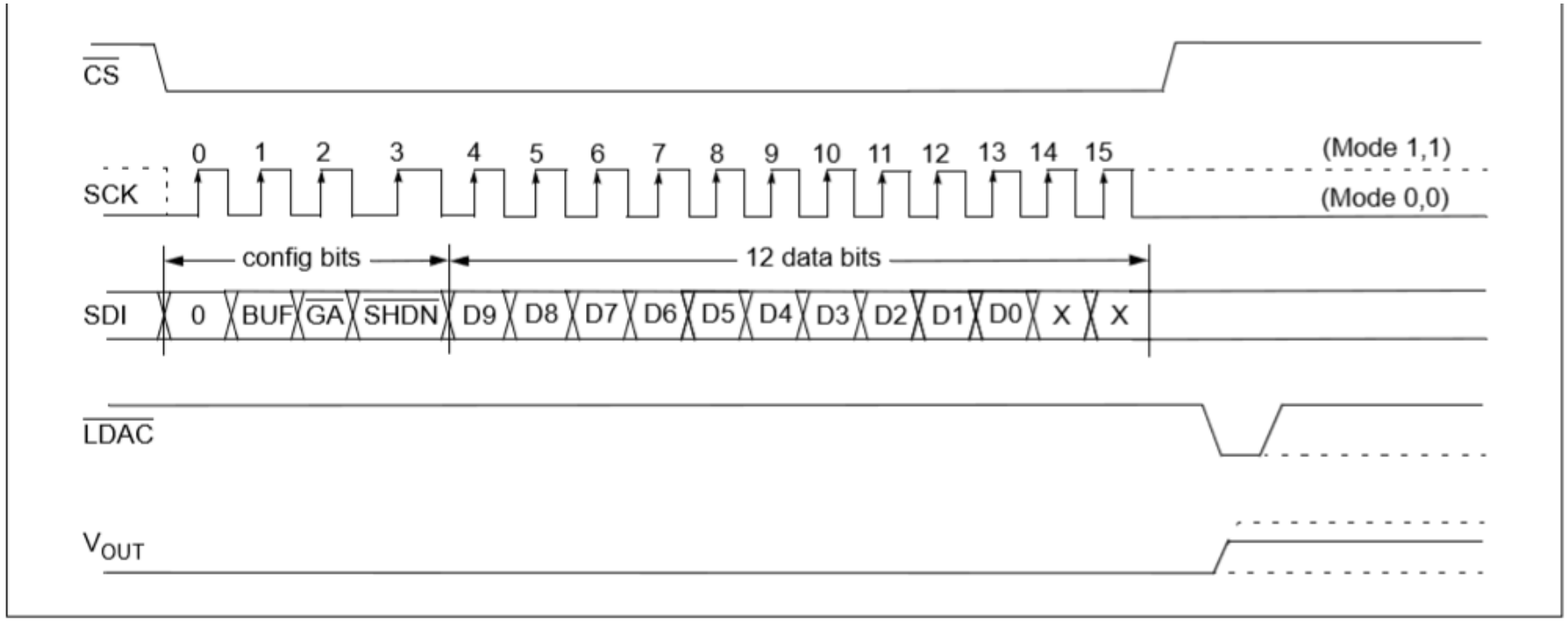


FIGURE 5-2: *Write Command for MCP4911 (10-bit DAC). Note: X are don't care bits.*

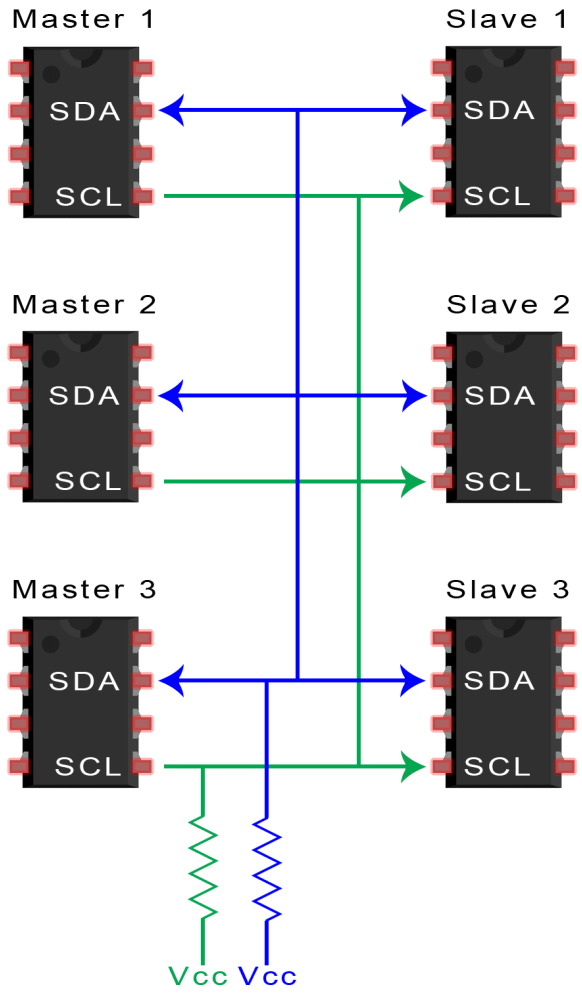
한동대학교 HANDONG GLOBAL UNIVERSITY

I2C (Inter-Integrated Circuit)

I2C (Inter-Integrated Circuit)

- A serial communication protocol
- Combined features of SPI and UART
  - Allow Multi-Master, Multi-Slave connection (SPI: single master)
  - Synchronous like SPI
  - Use 2 wires like UART
- Usually 100kbps, but can go as high as 5Mbps

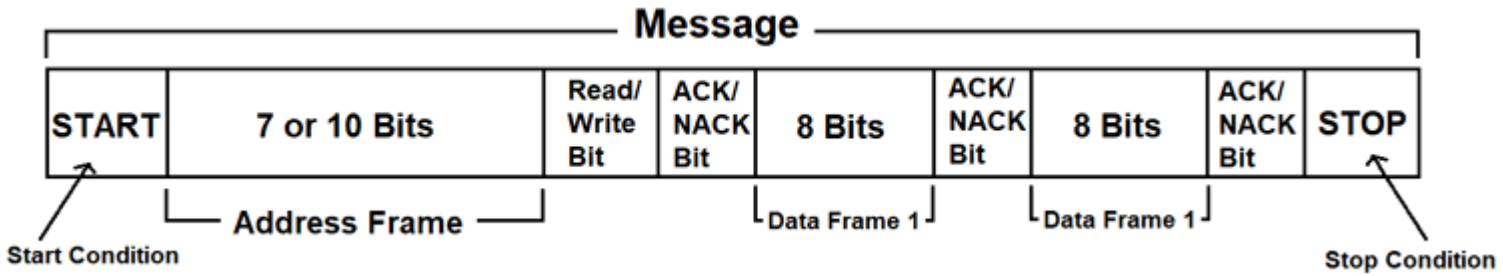
- SDA: Serial Data
  - Send/Receive data line
- SCL: Serial Clock
  - CLK signal



I2C (Inter-Integrated Circuit)

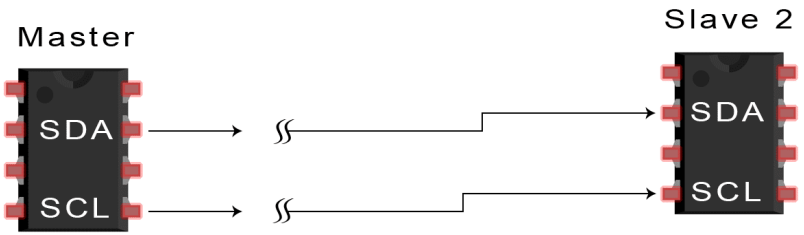
- **I2C (Inter-Integrated Circuit)**
  - Transfer data in messages: frames of data
  - Contains address of the slave, start, stop, read/write, ACK/NACK
  - Each message can contain several data frames(8bits)

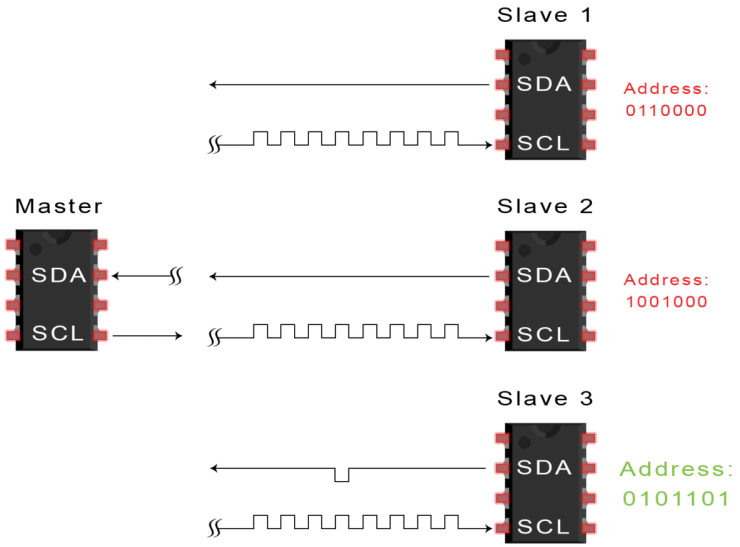
  - Start: 'H' → 'L' before SCL falling edge
  - Stop: 'L' → 'H' after SCL rising edge
  - Address: ID of slave
  - Read/Write: Master Sending('H'), Receiving ('L')
  - ACK/NACK: For each data frame received successfully, ACK is received

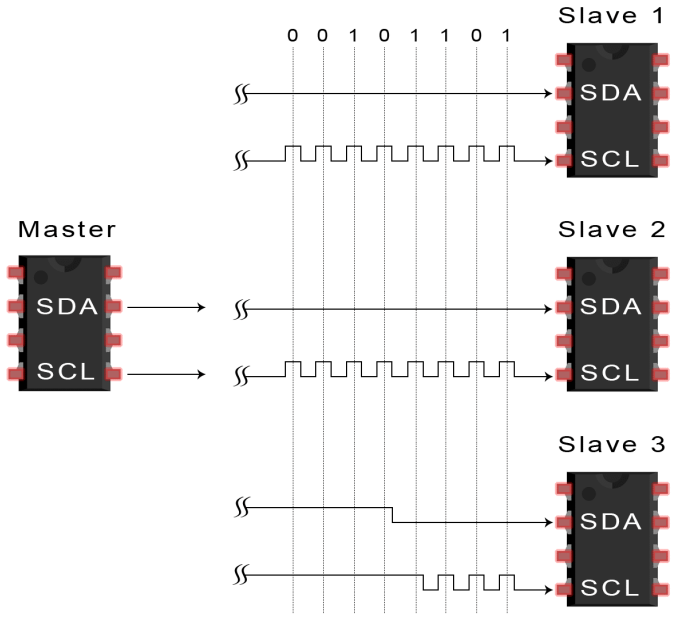


I2C (Inter-Integrated Circuit)

- **I2C (Inter-Integrated Circuit)**
  1) Start: 'H' → 'L' *before* SCL falling edge
  2) All slaves receive data (start+address +R/W bits)
  3) Matched Slave sends ACK bits (SDA 'H'->'L')
  4) Master sends or receives data frames
  5) Every data frame, slave sends ACK bits
  6) To Stop, 'L' to 'H' after switching SCL rising edge





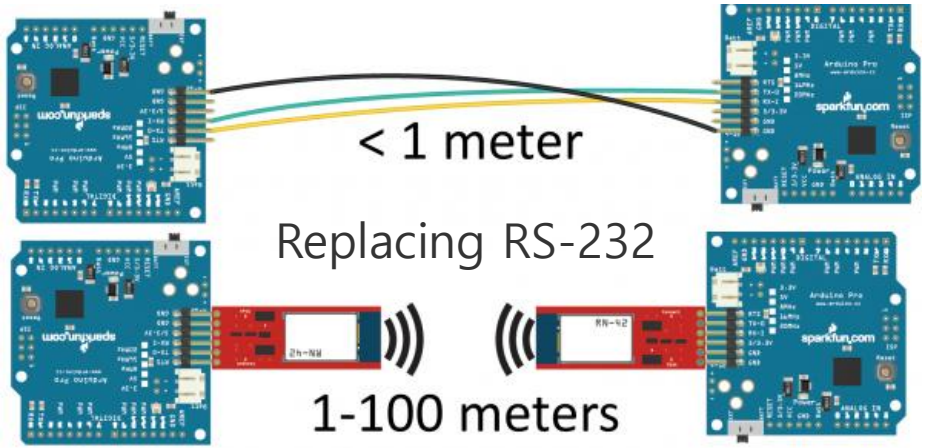


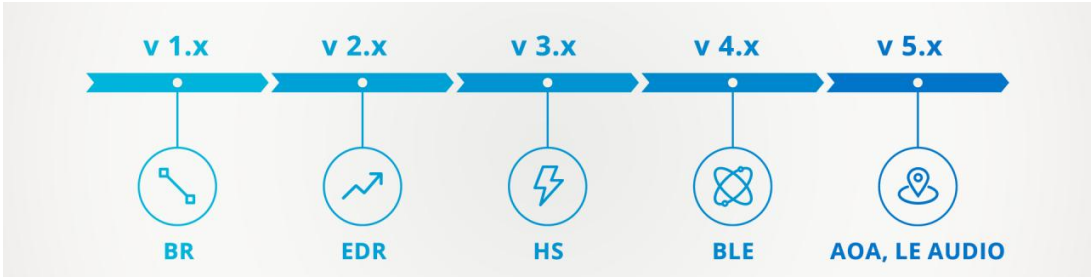
Bluetooth

- **Bluetooth**
  - A wireless serial communication protocol
  - Band 2.4GHz
  - unique 48-bit address (BD_ADDR)



MAC NO: 000666 **422152**





Version	4.0	4.1	4.2	5.0 & 5.1
Multi-Roles	No	Yes		
PDU Payload	Up to 31 bytes		Up to 255 bytes	
LE Secure	No		Yes	
IoT Support	Limited		Medium	High
Advertising Channels	3 Channels			3 Primary Ch. 37 Secondary Ch.
Data Rate	1 Mbps			2 Mbps
Effective Range	50 m (Line of Sight) 10 m (Indoor)			200 m (Line of Sight) 40 m (Indoor)
Battery Life	Shorter			Longer

Bluetooth

- **Bluetooth**
  - A wireless serial communication protocol using ISM: **Band 2.4GHz**
    - (Industrial, Scientific and Medical Frequency band)
  - Short range connectivity, license free spec.
  - Invented by Ericsson 1994, Maintained by SIG: Special Interest Group
  - Very low power. A master can be connected to several slaves

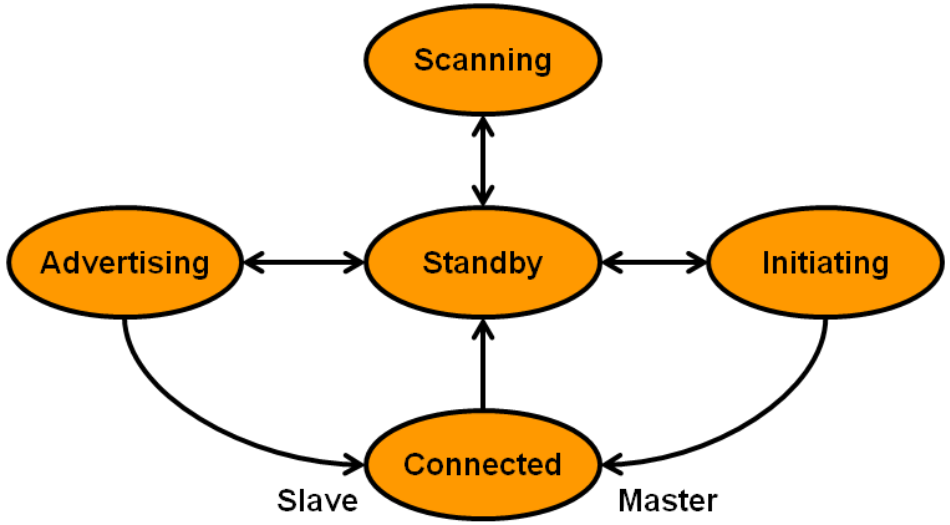
Standard	Description
Wifi	Up to 600 Mbits/sec Fixed wide frequency channels Requires lots of power Support for 2.4 and 5 GHz channels Extensive security features
Bluetooth/BLE	Very low power BT up to 2 Mbps Massive deployed base Frequency hopping Good performance in congested/noisy environment Ease of use, no roaming
ZigBee	Very low power Fixed channels Complex mesh network 250 kbps bandwidth

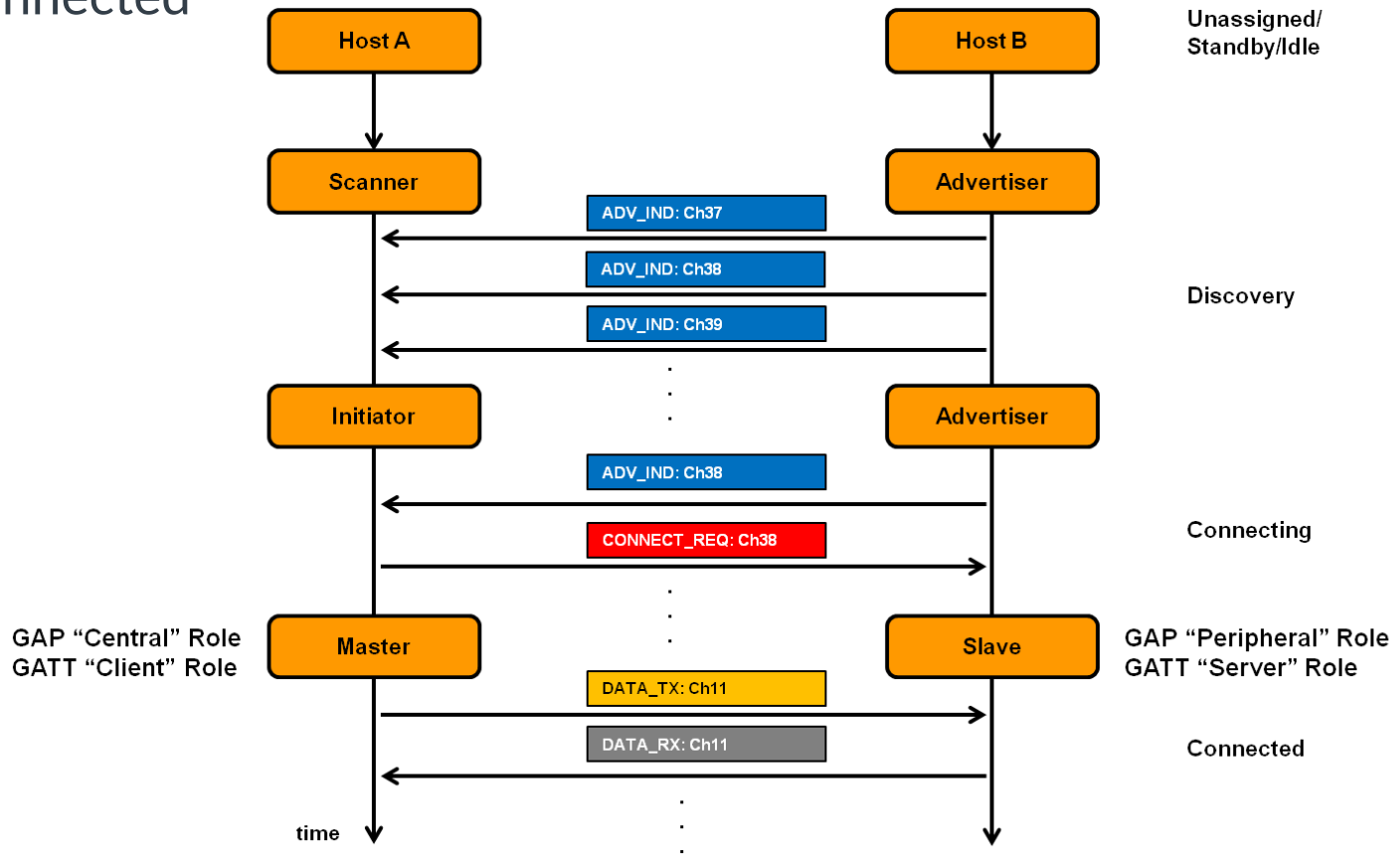
Bluetooth

- Bluetooth, Bluetooth Low Energy
  - Bluetooth: classical Bluetooth
    - Higher data rate: 1~3 Mbps
  - BLE (Bluetooth smart)
    - Designed focused for low powered
    - Lower data rate: ~1 Mbps
    - Not for audio transmission. Short data transmission
  - Bluetooth v4.0~: BLE & Bluetooth

Bluetooth

- BLE:
  - Exchange data in one of two states
    - Advertising and Connected modes
  - Standby -> Discovery -> Connecting -> Connected





https://microchipdeveloper.com/ble:start

Bluetooth

- Connecting Bluetooth module to MCU

